



Evaluation of the Gastrointestinal Patient

Full Review: May 2025 By [Stephanie M. Moleski](#), MD, Sidney Kimmel Medical College at Thomas Jefferson University | Peer reviewed by [Minhhuyen Nguyen](#), MD, Fox Chase Cancer Center, Temple University
Last updated: Jul 2025

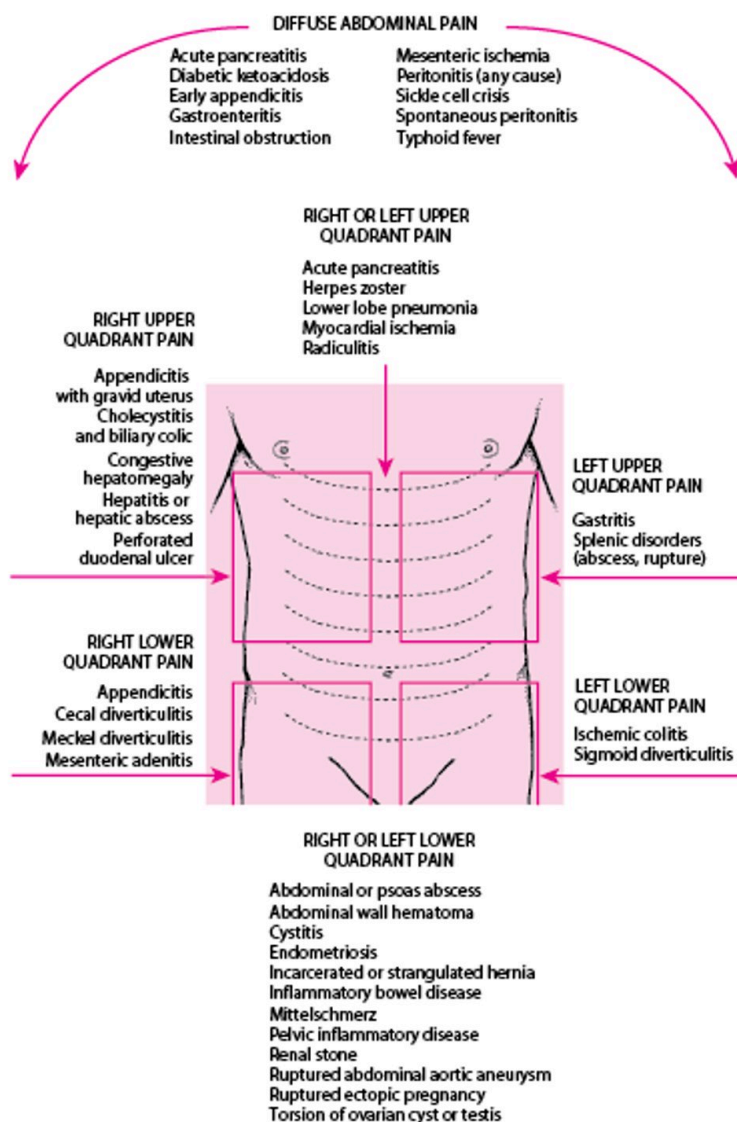
Gastrointestinal (GI) symptoms and disorders are quite common. History and physical examination are often adequate to make a diagnosis in some patients; in other cases, testing is necessary.

History in the GI Patient

Beginning with open-ended questions, the clinician identifies the location and quality of symptoms and any aggravating and alleviating factors.

Abdominal pain is a frequent GI complaint (see [Acute Abdominal Pain](#) and [Chronic and Recurrent Abdominal Pain](#)). Determining the location of the pain can help with the diagnosis. For example, pain in the epigastrium may reflect problems in the pancreas, stomach, or small bowel. Pain in the right upper quadrant may reflect problems in the liver, gallbladder, and bile ducts such as [cholecystitis](#) or [hepatitis](#). Pain in the right lower quadrant may indicate inflammation of the appendix, terminal ileum, or cecum, suggesting [appendicitis](#), ileitis, or [Crohn disease](#). Pain in the left lower quadrant may indicate [diverticulitis](#) or [constipation](#). Pain in either the left or right lower quadrant may indicate colitis, ileitis, or ovarian etiologies. (See figure [Location of Abdominal Pain and Possible Causes](#).)

Location of Abdominal Pain and Possible Causes



Asking patients about radiation of pain may help clarify the diagnosis. For example, pain radiating to the shoulder may reflect [cholecystitis](#) because the gallbladder may be irritating the diaphragm. Pain radiating to the back may reflect [pancreatitis](#). Asking patients to describe the character of the pain (ie, sharp and constant, waves of dull pain) and the onset (sudden, such as resulting from a perforated viscus or ruptured ectopic pregnancy) can help differentiate causes.

Patients should be queried about changes in eating and elimination, as well as difficulty swallowing ([dysphagia](#)), loss of appetite, and presence of [nausea and vomiting](#). If patients are vomiting, they should be asked how often and for how long and whether they have noted blood or coffee-ground-like material suggestive of [GI bleeding](#). Also, patients should be asked about the type and quantity of liquids they have tried to drink, if any, and whether they have been able to keep them down.

Patients should also be asked about elimination, including when their most recent bowel movement was, how frequently they have been having bowel movements, and whether this frequency represents a change from their typical frequency. It is more useful to ask for specific, quantitative information about

bowel movements rather than simply asking whether they are constipated or have diarrhea because different people use these terms quite differently. Patients should also be asked to describe the color and consistency of the stool, including whether stool has appeared black or bloody (suggestive of GI bleeding), purulent, or mucoid. Patients who have noticed blood should be asked whether it was coating the stool, mixed with stool, or whether blood was passed without any stool.

A [gynecologic history](#) is important in women because pregnancy or gynecologic and obstetric disorders may present with GI symptoms.

Associated nonspecific symptoms, such as [fever](#) or [weight loss](#), must be assessed. Unintentional weight loss is an associated symptom that may indicate a more severe problem such as cancer, and the clinician should be prompted to do a more extensive evaluation.

Patients report symptoms differently depending on their personality, the impact of the illness on their life, and sociocultural influences. For example, nausea and vomiting may be minimized or reported indirectly by a patient with severe depression but presented with urgency by a patient who is anxious.

Important elements of the past medical history include presence of previously diagnosed GI disorders, previous abdominal surgery, and use of medications and substances that might cause GI symptoms (eg, nonsteroidal anti-inflammatory drugs [NSAIDs], alcohol).

Physical Examination of the GI Patient

The physical examination might begin with inspection of the oropharynx to assess hydration, ulcers, or possible inflammation.

Inspection of the abdomen with the patient supine may show a convex appearance when bowel obstruction, ascites, or, rarely, a large mass is present. Auscultation to assess bowel sounds and determine presence of bruits should follow. Percussion elicits hyperresonance (tympany) in the presence of bowel obstruction and dullness with ascites and can determine the span of the liver. Palpation proceeds systematically, beginning gently to identify areas of tenderness and, if tolerated, palpating deeper to locate masses or organomegaly.

To assess the liver size, consistency, and tenderness, the examiner should place the examining hand on the right lower quadrant of the abdomen, parallel to the rectus muscle, and gently press upward while the patient takes a deep breath. The liver edge, if palpable, will descend and meet the fingertips. The examining hand is moved upwards, into the right upper quadrant and toward the inferior ribs, until the liver edge is palpated.

When the abdomen is tender, patients should be assessed for peritoneal signs such as guarding and rebound. Guarding is an involuntary contraction of the abdominal muscles that is slightly slower and more sustained than the rapid, voluntary flinch exhibited by sensitive or anxious patients. Rebound is a distinct flinch upon brisk withdrawal of the examiner's hand.

The inguinal area and all surgical scars should be palpated for [hernias](#).

Digital rectal examination (see [Evaluation of Anorectal Disorders](#)) with testing for occult blood and (in women) [pelvic examination](#) complete the evaluation of the abdomen.

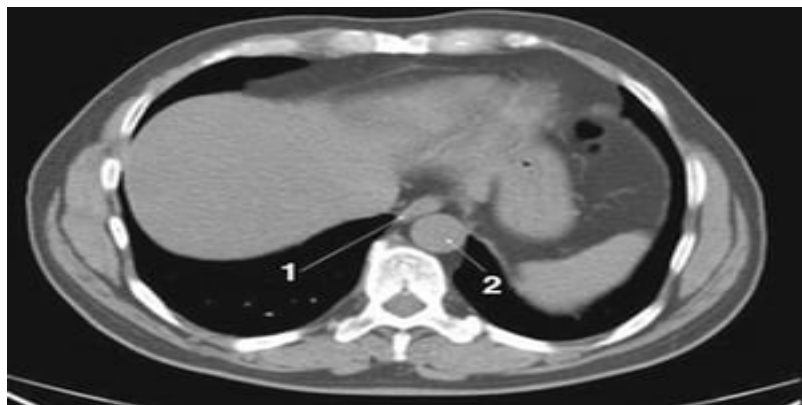
Testing in the GI Patient

Patients with acute, nonspecific symptoms (eg, dyspepsia, nausea) and an unremarkable physical examination rarely require testing. Findings suggesting significant disease (alarm symptoms) should prompt further evaluation:

- Anorexia
- Anemia
- Blood in stool (gross or occult)
- [Dysphagia](#)
- Fever
- Hepatomegaly
- Pain that awakens patient from sleep
- Persistent nausea and vomiting
- Weight loss

Chronic or recurrent symptoms, even with an unremarkable examination, also warrant evaluation. See also [Diagnostic and Therapeutic GI Procedures](#) for specific GI tests.

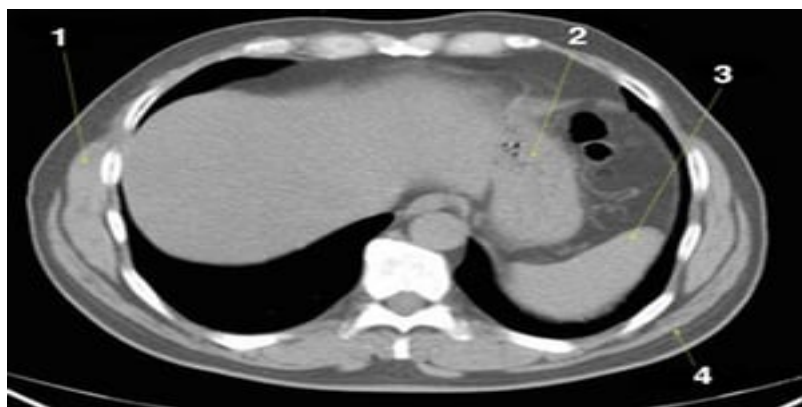
Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slides)



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 1)

1 = esophagus; 2 = aorta.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 2)

1 = serratus anterior; 2 = stomach; 3 = spleen; 4 = latissimus dorsi.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 3)

1 = liver; 2 = stomach; 3 = spleen.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 4)

1 = caudate lobe of liver; 2 = fissure for ligamentum venosum.

© Springer Science+Business Media

Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 5)

1 = fissure for ligamentum teres.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 6)

1 = inferior vena cava; 2 = aorta; 3 = portal vein.

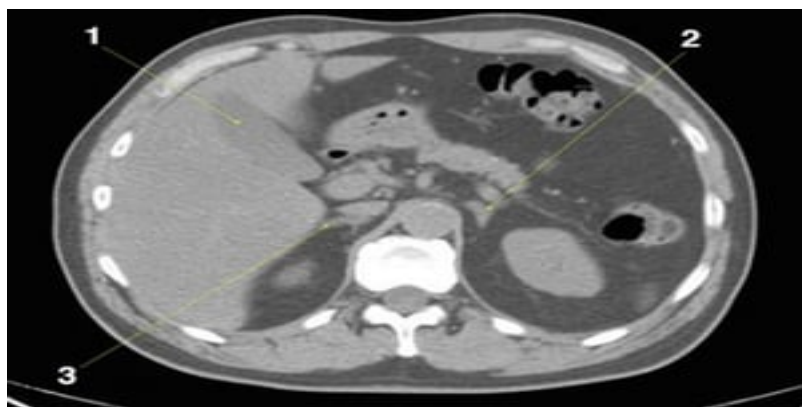
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 7)

1 = splenic vein.

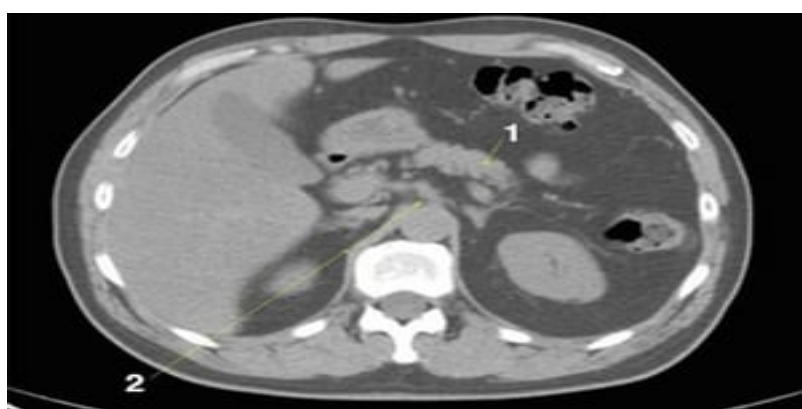
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 8)

1 = gallbladder; 2 = left adrenal gland; 3 = right adrenal gland.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 9)

1 = pancreas; 2 = celiac axis.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 10)

This image shows the crura of the diaphragm (arrows).

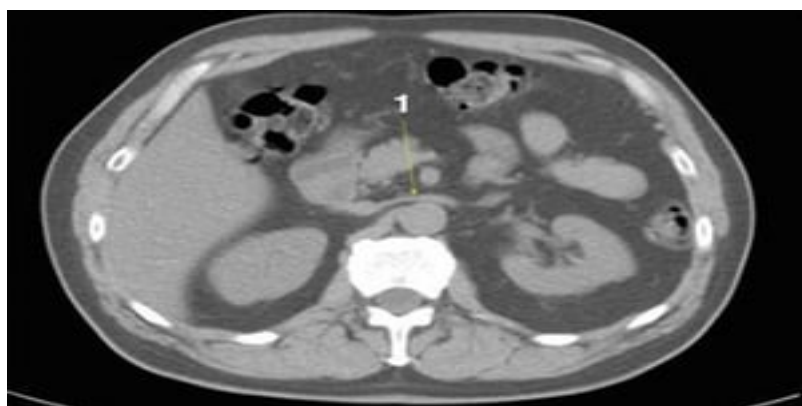
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 11)

1 = duodenum.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 12)

1 = left renal vein.

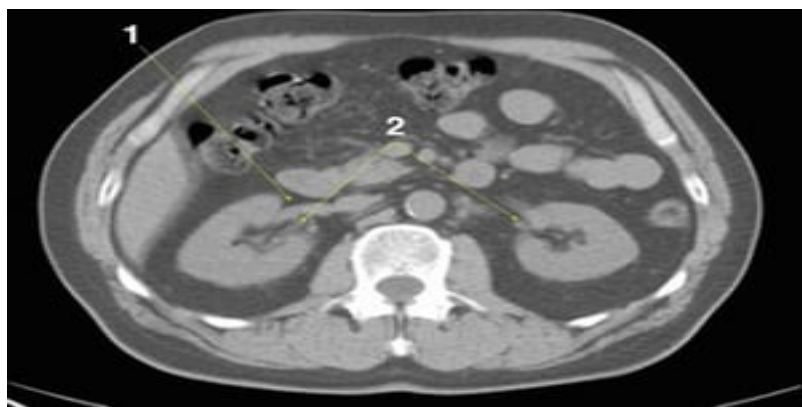
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 13)

1 = superior mesenteric vein; 2 = superior mesenteric artery.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 14)

1 = right renal vein; 2 = kidneys at hila.

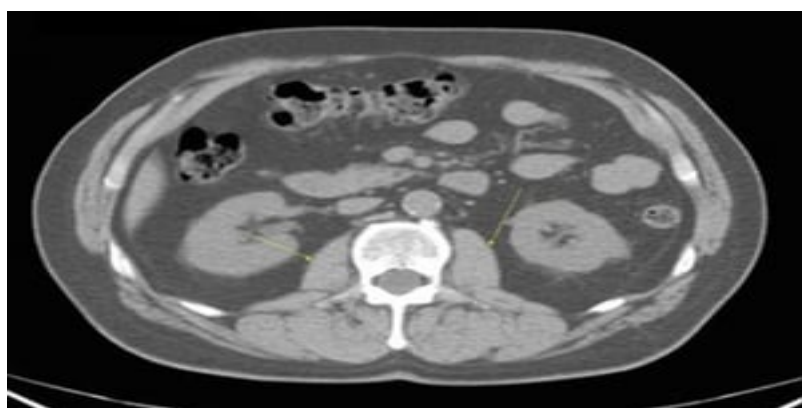
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 15)

1 = latissimus dorsi; 2 = serratus posterior.

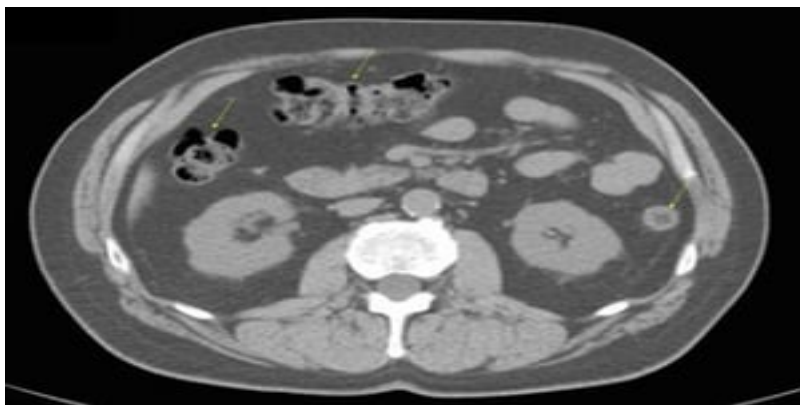
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 16)

This image shows the psoas muscles (arrows).

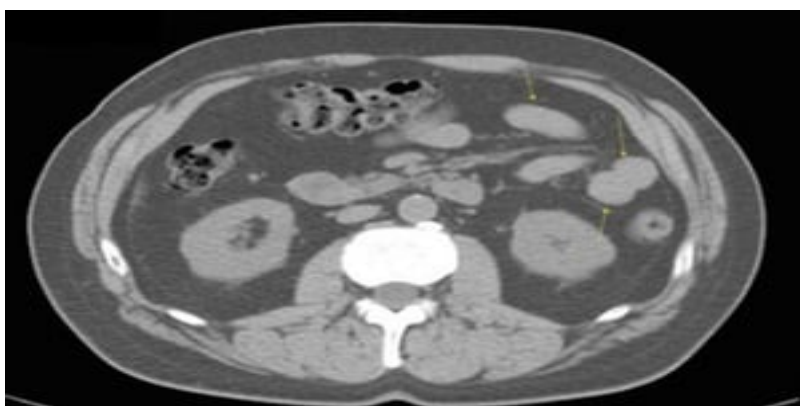
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 17)

This image shows the colon (arrows).

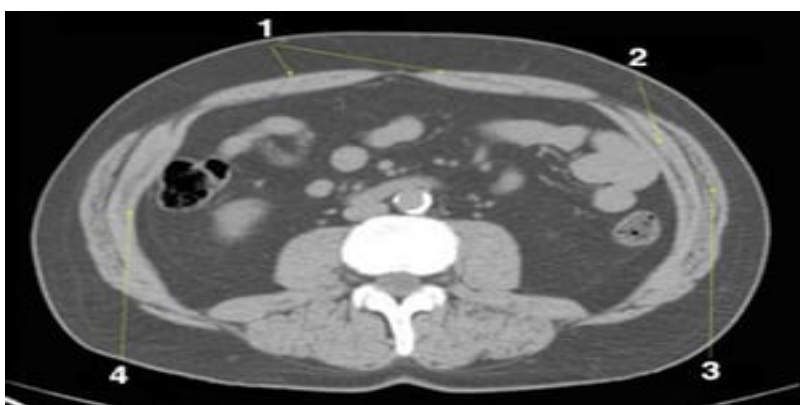
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 18)

This image shows the jejunum (arrows).

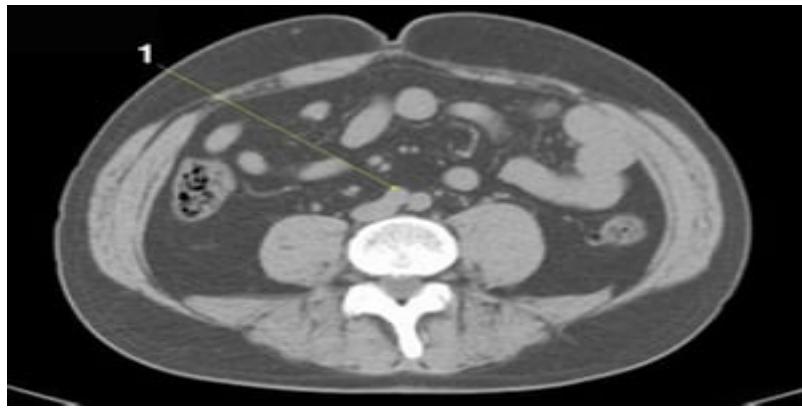
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 19)

1 = rectus abdominus; 2 = internal oblique; 3 = external oblique; 4 = transverse abdominus.

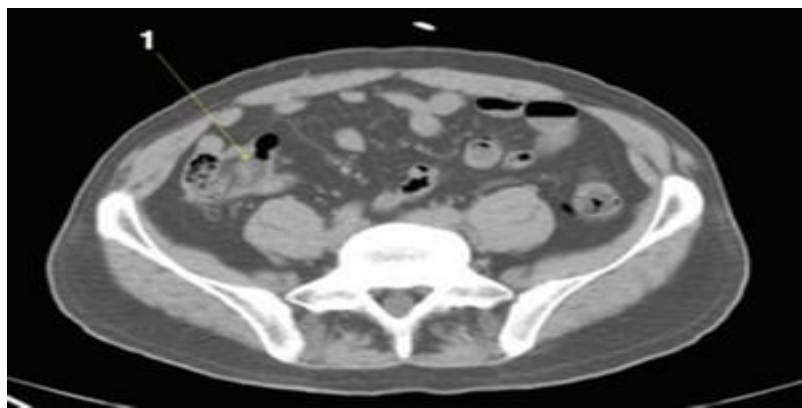
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 20)

1 = aortic bifurcation.

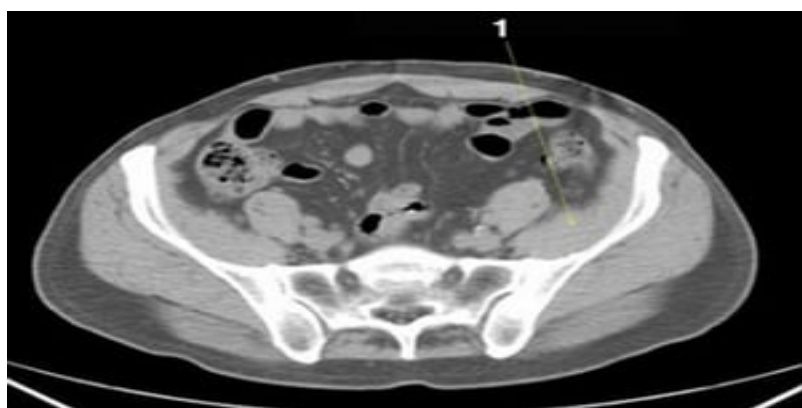
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 21)

1 = ileum.

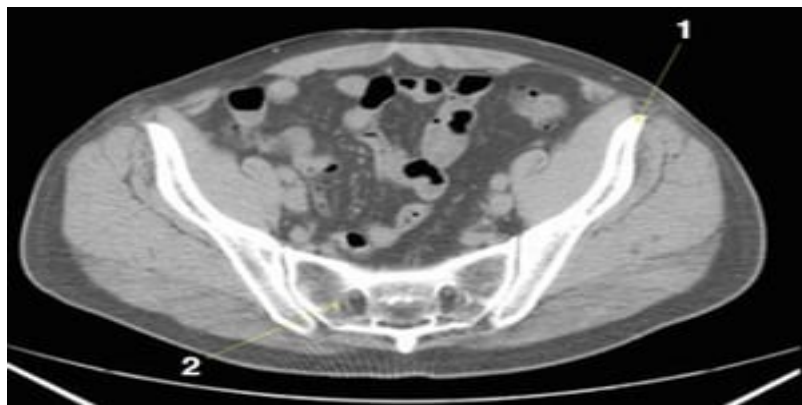
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 22)

1 = iliacus muscle.

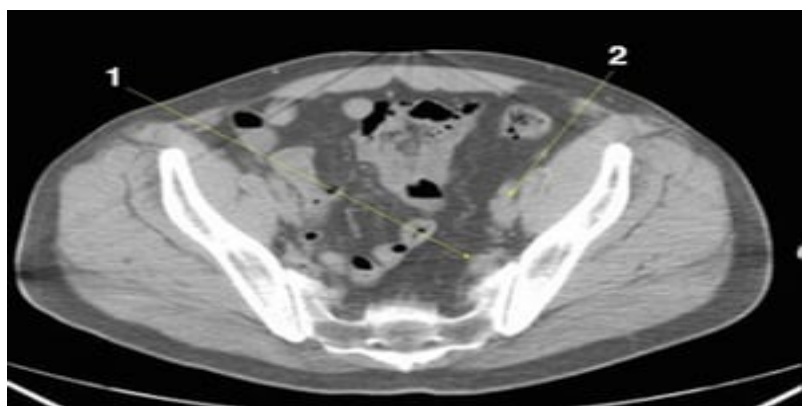
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 23)

1 = ilium; 2 = sacrum.

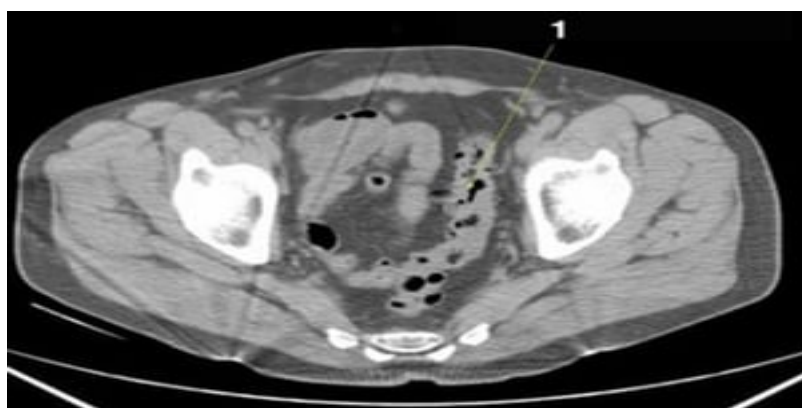
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 24)

1 = internal iliac vessels; 2 = external iliac vessels.

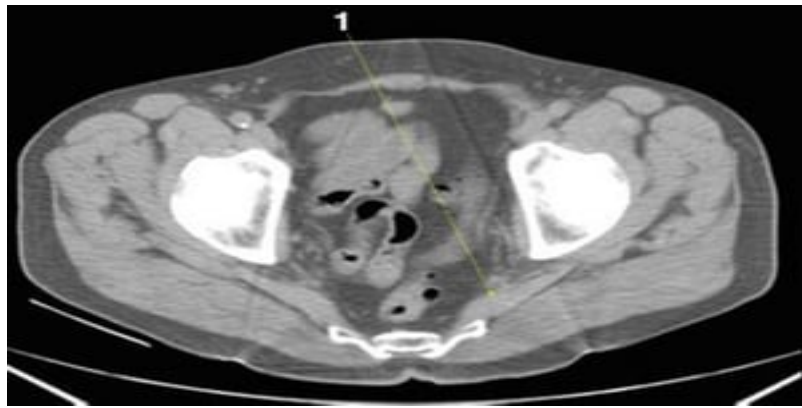
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 25)

1 = sigmoid colon.

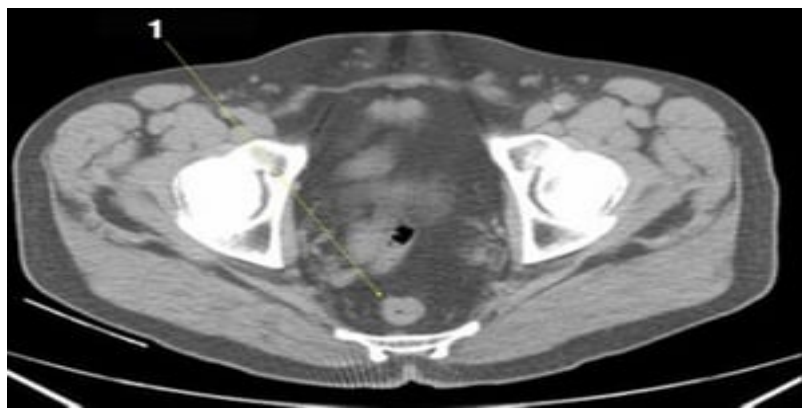
© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 26)

1 = piriformis.

© Springer Science+Business Media



Noncontrast CT Scan of the Abdomen and Pelvis Showing Normal Anatomy (Slide 27)

1 = rectum.

© Springer Science+Business Media



Copyright © 2026 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.